Erika Alexandra García Barrios

ML Engineer · Computer Vision · LLMs, RAG and Agents · Python

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PROFESSIONAL SUMMARY

Software Engineer focused on Machine Learning: I take models from experimentation to systems that are evaluated, monitored and actually work. In my thesis I compared ResNet50, EfficientNet-B3 and MobileViT-S to classify breast lesions in DCE-MRI with a patient-level split, calibration and Grad-CAM (ResNet50: ROC-AUC 0.941 and sensitivity 0.864 on 35 test patients); I presented it at XXI SIE–2026 and I am turning it into a paper. I build LLM, RAG and agent systems (LangGraph, MCP, Pinecone) evaluated against known-answer test sets, protected by more than 1,000 automated tests and served with FastAPI and Docker. I have a solid backend engineering foundation in Java and Python. Expected graduation: December 2026.

EXPERIENCE

AI Systems & Automation Engineer — Independent / Freelance — 2023 — Present

Design and development of AI systems with LLMs, RAG and agents, plus custom web development for clients, integrated through APIs, web interfaces and containers.

- SaberPro-RAG: RAG study assistant covering the five modules of Colombia's Saber Pro exam (Qdrant, sentence-transformers, Ollama / Mistral), documented with ADRs.
- Vella Perfumería (client): custom e-commerce with Next.js 16, React 19, TypeScript and Tailwind, and a Spring Boot 4 backend with Spring Security, JPA, Flyway and PostgreSQL; real catalog of more than 100 products, cart, payments and admin panel. Storefront live on Vercel.
- Montajes y Servicios García (client): bilingual (Spanish/English) landing page with light and dark themes, responsive design and direct WhatsApp contact, deployed on Vercel.

Python · FastAPI · React · Next.js · TypeScript · Spring Boot · PostgreSQL · Docker · LLMs · RAG · AI Agents · Redis · Vercel

Data Science & Machine Learning Researcher — CognIA Research Group — Institución Universitaria Pascual Bravo — Feb 2026 — Present

Predictive models to classify benign and malignant breast tumors with the Breast Cancer Wisconsin Diagnostic dataset.

- Compared logistic regression, decision trees, random forest, SVM and KNN with K-Fold cross-validation, stratified splits and hyperparameter tuning.

Python · Scikit-learn · Pandas · NumPy · Matplotlib · Seaborn · Transformers · CNN · SVM

AI & Computer Vision Researcher — Undergraduate thesis — Institución Universitaria Pascual Bravo — Sep 2025 — Jun 2026

Deep learning to classify benign and malignant breast lesions in DCE-MRI. Thesis approved in September 2026; now being turned into a scientific paper with the guidance of Dr. Rubén Darío Fonnegra Tarazona.

- Compared ResNet50, EfficientNet-B3 and MobileViT-S with transfer learning and progressive fine-tuning, K-fold cross-validation during training and a final patient-level hold-out evaluation (BreastDM).
- Final ResNet50 model on the test set (35 patients): ROC-AUC 0.941, sensitivity 0.864 (19 of 22 malignant), specificity 0.846 and F1 0.884; calibrated AUC (isotonic) 0.897, 95% CI [0.766, 0.992].
- Interpretability with Grad-CAM; experiment tracking with MLflow and TensorBoard.

Python · PyTorch · OpenCV · ResNet50 · EfficientNet-B3 · MobileViT-S · Grad-CAM · MLflow · TensorBoard

Webmaster · Institutional Web Development (contractor) — Institución Universitaria Pascual Bravo — Jul 2025 — Oct 2025

Management and development of the university's web platforms and support for digital transformation initiatives.

- Maintained, updated and optimized institutional web platforms, improving responsive design, usability and user experience.

HTML · CSS · JavaScript · Google Apps Script · WordPress · Moodle

Junior Full Stack Developer — Java Backend — Servisoft S.A. — Jun 2023 — Sep 2024

Custom Java backend solutions for public and private sector clients: DANE, JEP, Medellín City Hall, Medellín District Comptroller's Office, Bucaramanga Metropolitan Water Utility, UPB, Pascual Bravo and Viva.

- Backend development in Java 8/12 (Java SE, JSP, Servlets, JDBC) on Apache Tomcat 6–9, with REST and SOAP APIs for critical applications.

Java 8/12 · JSP · Servlets · JDBC · Apache Tomcat · REST · SOAP · MSSQL · Oracle · MongoDB · Git

Software Development Intern — MV-TEL SAS — Jun 2021 — Nov 2021

Full development cycle of a mobile app and the corporate website within a Scrum team.

- Requirements gathering and analysis, functional design, architecture definition and feature validation.

Java · Selenium WebDriver · HTML · CSS · JavaScript · Scrum

RESEARCH AND PUBLICATIONS

- Scientific paper (in preparation): deep learning classification of benign and malignant breast lesions in DCE-MRI (ResNet50, EfficientNet-B3 and MobileViT-S; ROC-AUC 0.941), derived from the approved thesis, under the guidance of Dr. Rubén Darío Fonnegra Tarazona.
- Speaker at the 21st International Energy Symposium — Expotecnológica 2026 (XXI SIE–2026): “Breast lesion classification using deep learning on magnetic resonance imaging”, co-authored with Rubén Darío Fonnegra.

SELECTED PROJECTS

Breast Cancer Classification in DCE-MRI — Thesis

Compared ResNet50, EfficientNet-B3 and MobileViT-S with patient-level evaluation, ROC curves and Grad-CAM. ROC-AUC 0.941.

JARVIS — Personal AI Assistant Platform

Modular AI system with a FastAPI backend and a React dashboard.

- LLMs with automatic fallback: a 6-model chain (GPT-OSS 120B/20B and Qwen on Groq, Gemini, local Mistral and Phi-3 with Ollama) behind an in-house gateway with token control, retries and circuit breaker.
- Orchestrator with 4 specialized agents (developer, security, reviewer and drafter), function calling, risk-based permissions, human approval and an audit log of every tool call.
- RAG on Pinecone with multilingual-e5 embeddings and a BGE reranker, validated with a custom evaluation set (recall@3 = 1.0).
- Observability across 9 Docker Compose services (backend, frontend, Redis, Career Ops, Prometheus, Grafana, Loki, Promtail and Jaeger): backend and Career Ops metrics in Prometheus with 6 alerts (service down, 5xx errors, failing tools, action awaiting approval and LLM rate limit), Grafana dashboards, centralized logs with Loki and Promtail, and distributed tracing with Jaeger.

FastAPI · React · Claude · Groq · Ollama · Pinecone · Redis · Supabase · MLflow · Prometheus · Grafana · Docker · Kubernetes · Loki · Jaeger · Docker Compose

Lector de PDF — on-device AI inference

milectorpdf.vercel.app

Live PWA that turns PDFs and books into audiobooks with AI models running 100% in the user's browser: no server, no inference cost and the document never leaves the device.

- Text-to-speech with Piper (VITS) ONNX models running on ONNX Runtime Web (WASM or WebGPU) inside a Web Worker, espeak-ng phonemization in WebAssembly and sentence-level synthesis with prefetching for continuous reading.
- Custom Spanish text-normalization pipeline (numbers, abbreviations, acronyms, OCR errors) and local OCR with Tesseract.js.
- Prosody correction: measured that questions did not rise in pitch (F0 via autocorrelation) and applied a pitch contour with WSOLA; intelligibility evaluated with Whisper.
- Offline neural translation (Firefox Translations Marian models, 8-bit quantized) in WebAssembly; 601 unit tests and 57 end-to-end tests.

ONNX Runtime Web · Piper / VITS · WebGPU · WebAssembly · Tesseract.js · Whisper · Marian NMT · TypeScript · PWA

Breast Cancer Diagnosis Support Tool — Project Lead

Five-person team project for the Intermediate AI Diploma at Universidad de Antioquia (2026), which I led: 10 supervised classifiers on Breast Cancer Wisconsin Diagnostic held to a clinical standard of recall ≥ 0.90 for the malignant class, deployed on AWS.

- Coordinated the team and the repository: two models per member, shared metrics (accuracy, precision, recall, F1, ROC-AUC, false negatives and positives) and consolidated results.
- Author of the EDA that selected 6 features for discriminative power and low multicollinearity, and of the comparative review of the 10 models.
- My models: Random Forest and SVM with recall-optimized 5-fold grid search and threshold tuning. Random Forest on 114 test cases: accuracy 0.98, recall 0.95, precision 1.00 and ROC-AUC 0.996 (2 false negatives, 0 false positives).
- Cloud deployment: Docker containers on AWS (EC2), an isolated VPC network, IAM permissions and image storage on S3.

Python · Scikit-learn · Random Forest · SVM · GridSearchCV · Pandas · AWS · EC2 · S3 · VPC · IAM · Docker

TECHNICAL SKILLS

Machine Learning & Deep Learning: Python, PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, Computer vision, Medical imaging, CNN, RNN, Transformers, Transfer learning, Grad-CAM

Generative AI & agents: LLMs, Generative AI, Claude, GPT-OSS, Llama 3, Mistral 7B, Phi-3, RAG, Embeddings, Semantic search, Reranking, Pinecone, Qdrant, Prompt engineering, AI agents, Multi-agent systems, Function calling, Local inference with Ollama, Whisper, Claude Code

MLOps, cloud & infrastructure: MLflow, Databricks, PySpark, Apache Spark, Terraform, TensorBoard, Hadoop, Airflow, ETL pipelines, Docker, Kubernetes, AWS (EC2, S3, VPC, RDS), GitHub Actions, Prometheus, Grafana, Vercel

Classic ML models: Linear regression, Logistic regression, KNN, Decision trees, Random Forest, SVM, Naive Bayes, K-Means, Gradient Boosting

Data science: Pandas, NumPy, Matplotlib, Seaborn, Jupyter, Time series, Multivariate regression, Fuzzy logic, Monte Carlo simulation, Statistical analysis, Anomaly detection, Power BI, Risk analysis, Parquet

EDUCATION

B.Sc. in Software Engineering — Institución Universitaria Pascual Bravo — Feb 2025 — Dec 2026

Coursework completed; graduation expected in December 2026.

Artificial Intelligence Diploma · Intermediate (280 h) — Universidad de Antioquia + Talento Tech — Feb 2026 — Apr 2026

Cloud Architecture Diploma · Advanced (232 h) — ESUMER — Oct 2024 — Dec 2024

Web Programming — Universidad Pontificia Bolivariana — Oct 2022 — Dec 2022

Data Science Diploma (380 h) — ESAP — Aug 2022 — Oct 2022

Software Development Technologist — Institución Universitaria Pascual Bravo — Aug 2017 — Jan 2022

Computer Maintenance Technician — SENA — 2017 — 2018

Process Automation Specialization (36 h) — Nodo + Accenture — Sep 2026 — Nov 2026

In progress: n8n, Zapier, Make, LLM APIs and RAG chatbots.

Digital Transformation with AI and Automation (certified) — Nodo + Accenture Scholarships — 2026 · cohort G3

AI and automation applied to digital transformation: process analysis and identification of automation opportunities.

All of my education has been funded by scholarships earned through good academic performance: Sapiencia, Participatory Budget, Matrícula Cero, Nodo + Accenture and Estud-IA.

CERTIFICATIONS AND AWARDS

- COPNIA professional license — Professional Registration Certificate (Auxiliary).
- 3rd place — Advanced Cloud Architecture Hackathon, ESUMER (Dec 2024): AWS infrastructure with Terraform.

LANGUAGES

Spanish — native · English — B1+ (intermediate)

EARLIER EXPERIENCE · TECHNICAL SUPPORT AND CUSTOMER SERVICE

Multi-Skill Agent — Technical support, sales and retention — OneLink BPO — Aug 2019 — Mar 2021

Tier-1 support for internet, TV and phone services (HFC, DTH); tested RPA bots in a test environment, reporting and documenting defects.

Technical Support Representative — Emtelco — Jan 2018 — Jan 2019

Troubleshooting of fiber (FTTH), HFC and UMTS services; equipment setup and connectivity testing.

IT Maintenance Intern — Espumas Medellín — May 2017 — Jan 2018

Support for 100+ internal users, operating system installation and backups, technical manuals and user training.

Customer Service Representative — Teleperformance — Mar 2014 — Nov 2014

Remote support for customers in Spain: billing, complaints and fiber service support.